Samy Youssoufine

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EDUCATION

EMINES – University Mohammed VI Polytechnic (UM6P) 2025 – 2030
Engineering in Industrial Management (Second year of Integrated Preparatory Classes) Benguerir, Morocco

- **Clubs**: Workshop Lead at **E++** (CS) and **E-Physics**; Sponsoring Lead at **DataLab**; Event Organizer at **E-Vox**.

Lycée Scientifique Al Hanane II 2022 – 2025
Baccalauréat in Mathematical Sciences (Engineering Option, BIOF) – High Honors Agadir, Morocco

- **Founder & President, PixelLab**: Founded the 3rd Hack Club in Morocco; taught weekly coding workshops for students.

TECHNICAL SKILLS

Hardware & CAD: Arduino (Uno, Mega 2560), C/C++ Firmware, Stepper Drivers, H-Bridges, PID / Cascaded Control, Quadrature Encoders, UART / Bluetooth (HC-06), WebSerial API, SolidWorks CAD, 3D Printing (PLA), CNC Milling

Software & Web: Python (NumPy, SciPy, Pandas, Matplotlib, Pygame, Flask), JavaScript (ES6+, Web Workers, HTML5 Canvas), SQL (SQLite), HTML/CSS, Git, Docker, Linux, GitHub Actions (CI/CD), MkDocs

Visual Design: UI/UX Prototyping, Adobe Illustrator, Photoshop, Premiere Pro, DaVinci Resolve, LaTeX / TikZ

HARDWARE & SOFTWARE PROJECTS

3-DOF CoreXY CNC Plotter Code | Live HMI | Report | *Arduino Mega, C++, JavaScript, SolidWorks* Feb – Jun 2026

- Built a 3-axis CNC pen plotter (300×300 mm area) achieving **0.2 mm precision**, earning the **top grade in the class**, and authoring an 85-page technical report.
- Wrote C++ firmware from scratch for synchronized stepper control, smooth speed ramping, and limit-switch safety via an integer Digital Differential Analyzer (DDA).
- Built a **browser-based control panel** with live preview and toolpath optimisation, with no external libraries and automated CI/CD deployment via GitHub Actions.

Autonomous 3-Wheeled Mobile Robot Code | SysID | Report | *Arduino Uno, PID, SolidWorks, Python* Sep – Dec 2025

- Designed and built an autonomous tricycle robot capable of carrying a **15 kg payload** from CAD to final wiring, earning the second highest grade in class and authoring a 73-page technical report.
- Performed experimental DC motor system identification to tune a closed-loop **PID speed and position controller**.
- Created a **WebSerial and Bluetooth (HC-06)** web dashboard with an interactive 2D canvas for point-and-click navigation.

E-Physics Simulation Suite Guide | *Python, Pygame, Jupyter, HTML/CSS* Oct 2026

- Co-developed a zero-install physics toolkit bundling portable Python environments and interactive simulations across 4 OS platforms.

P.W.P.-6: Autonomous Aquatic Cleaning Robot | *Power Systems, Hardware Architecture, Computer Vision* Aug 2026 – Present

- Sizing the solar panels and battery, and designing trash detection with computer vision for an autonomous cleaning boat.

Na't: School Application Platform (Orient'ino Souss Hackathon) | *Python, Flask, SQLite, JavaScript* Apr 2025

- Led product design, backend routing, SQLite schema modeling, and live pitch for a centralized student-school application portal; **won 2nd place out of 20+ teams**, including university teams.

HONORS & LEADERSHIP

- **Jury Prize: Hack Club Scrapyard Casablanca (Mar 2025)**: Built a web arcade with 2 teammates; won the **Lenovo** Jury Prize.
- **Head of Documentation & Archives: ULC TED-Ed (2026 – Present)**: Managing internal documentation and reports, chronicled 4 masterclasses, and co-organizing the inter-department Talkathon competition.
- **Best Delegate: EMC Youth MUN (Apr 2026)** Position Paper: Won top committee award representing China in UNGA on digital data sovereignty.
- **4th place: UM6P Robotics Competition (Feb 2026)**: Competed against 15 teams from all departments of UM6P to control a bipedal robot, and was selected to represent UM6P at the **National Robotics Competition (Mar 2026)**.
- **Open-Source LaTeX Lecture Notes** GitHub: Typeset and published complete first-year university math lecture notes.